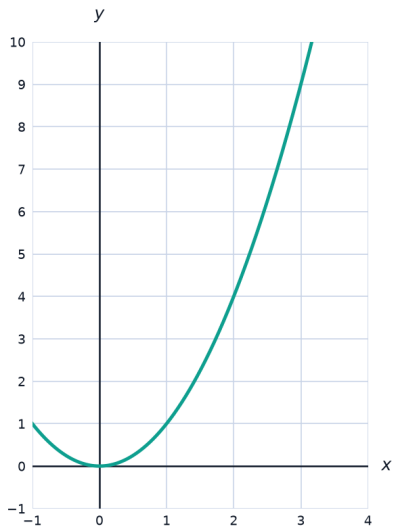
Integration & Applications

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Area under a curve

Find the area under $y = x^2$ from $x = 0$ to $x = 3$, using a definite integral.



2 [Foundational] • Target: Area under a curve

Find the area under $y = 4 - x^2$ from $x = -2$ to $x = 2$.

3 [Foundational] • Target: Signed area and regions below the axis

Evaluate $\int_0^2 (x - 2)\,dx$ and explain what the sign of the result means.

4 [Foundational] • Target: Signed area and regions below the axis

Find the (positive) area between $y = x - 2$ and the x-axis, from $x = 0$ to $x = 2$.

5 [Foundational] • Target: Finding a function from its derivative

A function satisfies $f'(x) = 6x^2 - 4$ and $f(1) = 5$. Find $f(x)$.

6 [Foundational] • Target: Average value of a function

Find the average value of $f(x) = x^2$ on $[0, 3]$, using $\text{avg} = \frac{1}{b-a}\int_a^b f(x)\,dx$.

7 [Foundational] • Target: Average value of a function

Find the average value of $f(x) = 2x + 1$ on $[1, 5]$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Word problems: integration in context

This question has two parts. A particle moves along a line with velocity $v(t) = 6t - 3t^2$ (m/s) for $0 \le t \le 4$.

a Find the particle's displacement from $t = 0$ to $t = 2$, using $\int_0^2 v(t)\,dt$.

b Find the particle's displacement from $t = 0$ to $t = 4$, and explain why it is not simply twice the part (a) answer.

9 [Intermediate] • Target: Word problems: integration in context

This question has two parts. Water flows into a tank at a rate of $r(t) = 20 - 2t$ liters per minute, for $0 \le t \le 10$.

a Write a definite integral for the total volume of water that flows in during the first 5 minutes.

b Evaluate the integral to find the volume.

10 [Intermediate] • Target: Finding a function from its derivative

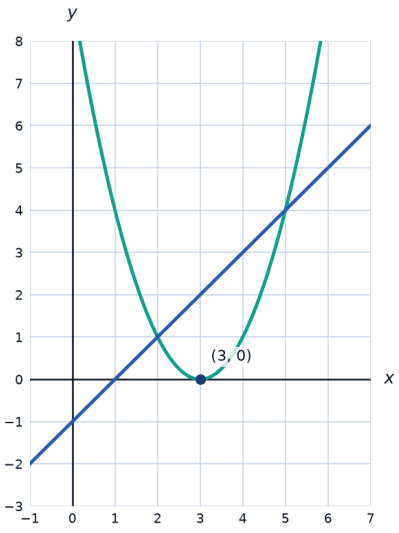
Find the area under the line $y = x$ above the x-axis from $x = 0$ to $x = 4$.

11 [Intermediate] • Target: Average value of a function

Evaluate the definite integral: $\int_0^3 (2x + 1)\,dx$

12 [Intermediate] • Target: Word problems: integration in context

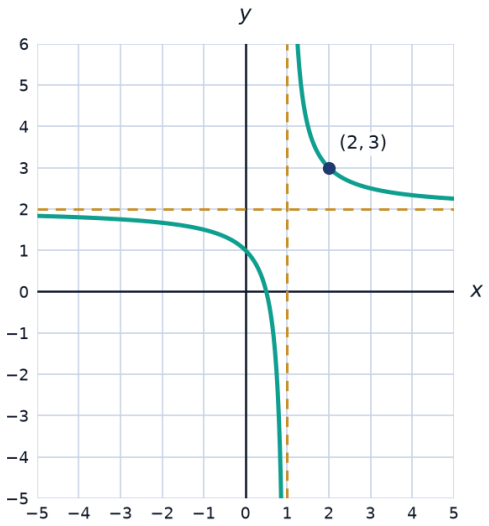
The function $f(x) = x^2 - 6x + k$ represents a parabola that touches the x-axis at $x = 3$. What is the value of k?



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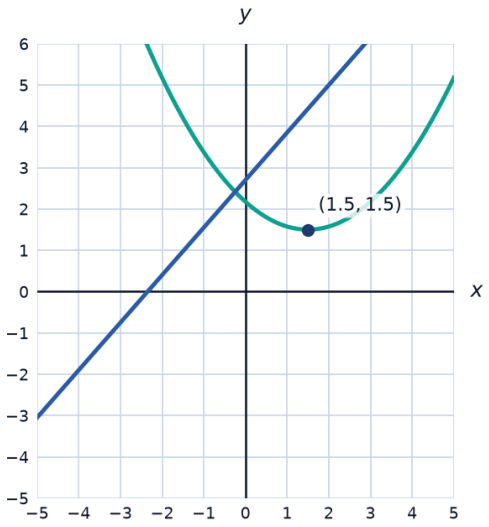
13 [Intermediate] • Target: MCQ — Basic antiderivatives (the power rule)

The graph of the function $f(x)$ is shown below. What is the value of $\lim_{x \to 2} f(x)$?



14 [Intermediate] • Target: MCQ — Antiderivatives of polynomials

In the diagram below, the straight line $y = mx + c$ intersects the x and y axes. If its slope is $\frac{1}{2}$ and it passes through the point $(4, 5)$, find the value of the constant term c.



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: MCQ — Antiderivatives of trigonometric and exponential functions • Cross-Lesson: 4.01 Trigonometry

If $\int_1^4 f(x)\,dx = 7$ and $\int_2^4 f(x)\,dx = 3$, determine the value of $\int_1^2 f(x)\,dx$.

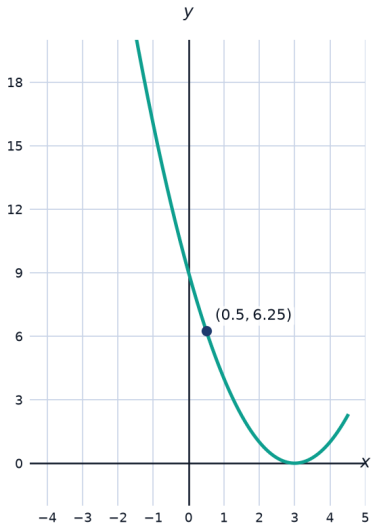
16 [Challenging] • Target: MCQ — Evaluating definite integrals • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Find the general indefinite integral: $\int (4x^3 - 6x + 5)\,dx$

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17 **[Challenging]** • **Target: MCQ — Definite integrals with polynomial integrands** • **Cross-Lesson: 6.01 Limits & Derivatives**

The function $f(x) = x^2 - 6x + 9$ represents the trajectory of a projectile. What is the minimum height the object reaches?

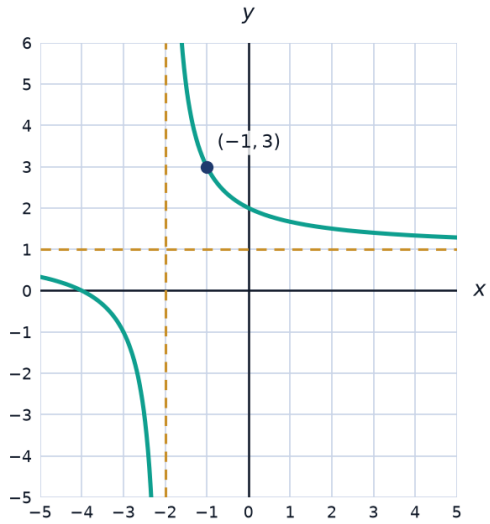


18 **[Challenging]** • **Target: MCQ — Definite integrals of trigonometric and exponential functions** • **Cross-Lesson: 8.01 Exponentials & Logarithms**

Level 3 · Advanced Find the exact surface area generated by revolving the line segment $y = 2x$ on $[0, 3]$ about the x-axis.

19 **[Challenging]** • **Target: MCQ — Area under a curve** • **Cross-Lesson: 9.01 Statistics & Probability**

The graph of the function $f(x)$ is shown. What is the limit of $f(x)$ as x approaches 2?



20 **[Challenging]** • **Target: MCQ — Signed area and regions below the axis** • **Cross-Lesson: 1.01 Algebra & Equations**

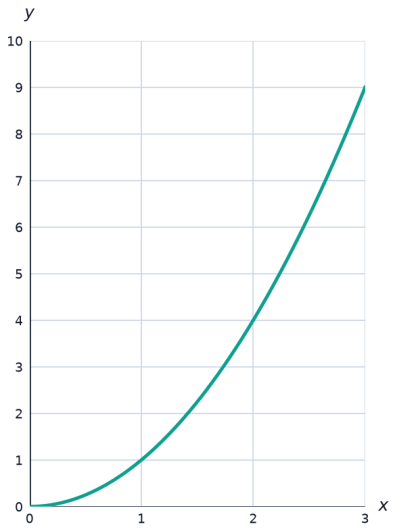
A particle is projected vertically upwards. The function $f(t) = -5t^2 + 20t + 1$ represents the height (in meters) of the particle after t seconds. What is the maximum height reached by the particle?

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 **[Foundational]** • **Target: MCQ — Area under a curve**

The graph shows $y = x^2$. Find the area under the curve from $x = 0$ to $x = 3$, using a definite integral.



a. 9

b. 18

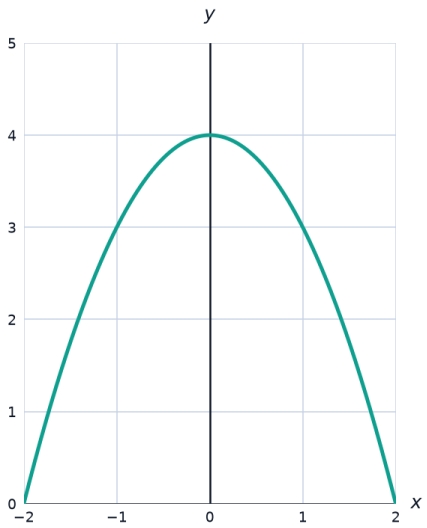
c. 3

d. 27

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22 [Foundational] • Target: MCQ — Area under a curve

The graph shows $y = 4 - x^2$. Find the area under the curve from $x = -2$ to $x = 2$.



a. $\frac{16}{3}$

b. 32

c. 16

d. $\frac{32}{3}$

23 [Foundational] • Target: MCQ — Signed area and regions below the axis

Evaluate $\int_0^4 (x - 4)\,dx$ and state whether the region is above or below the x-axis.

a. 8; above

b. -8; above

c. 8; below

d. -8; below

24 [Foundational] • Target: MCQ — Word problems: integration in context

A particle moves along a line with velocity $v(t) = 8t - 2t^2$ (m/s) for $0 \le t \le 4$. Find the particle's displacement over this interval.

a. 64

b. $\frac{64}{3}$

c. 32

d. $\frac{192}{3}$

25 [Foundational] • Target: MCQ — Word problems: integration in context

Water flows into a tank at a rate of $r(t) = 30 - 3t$ liters per minute, for $0 \le t \le 10$. Find the total volume of water that flows in during this time.

a. 300

b. 150

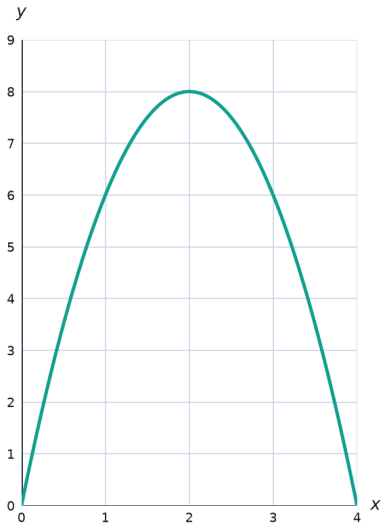
c. 450

d. 100

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26 [Foundational] • Target: MCQ — Word problems: integration in context

The graph shows a particle's velocity $v(t) = 8t - 2t^2$ for $0 \le t \le 4$. Find the particle's displacement (the area under the curve).



a. $\frac{64}{3}$

b. 16

c. 32

d. 64

27 [Foundational] • Target: MCQ — Word problems: integration in context

A factory's production rate is $p(t) = 50 + 4t$ items per hour, for $0 \le t \le 8$. Find the total number of items produced during this 8-hour period.

a. 528

b. 656

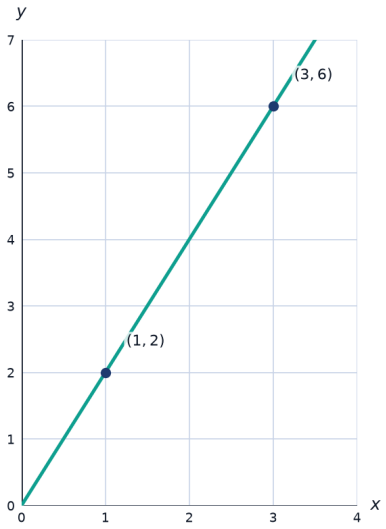
c. 450

d. 400

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28 [Foundational] • Target: Signed area and regions below the axis

[redacted] $f(x) = 2x$ [redacted] $x = 1$ [redacted] $x = 3$[redacted] [redacted] [redacted] [redacted]. To find the area under the curve of the function $f(x) = 2x$ from $x = 1$ to $x = 3$, we calculate the definite integral.



a. 4

b. 6

c. 8

d. 10

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Properties of definite integrals

[redacted] $f(x) = 2x$ [redacted] $x = 0$ [redacted] $x = 3$[redacted] [redacted] What is the numerical value of the definite integral of the function $f(x) = 2x$ from $x = 0$ to $x = 3$, representing the area under the curve?

a. 6

b. 9

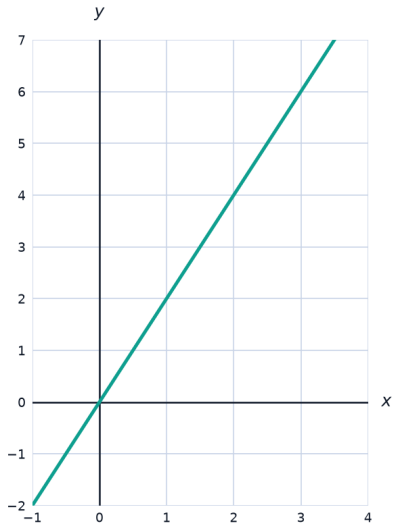
c. 12

d. 3

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30 [Intermediate] • Target: Finding a function from its derivative

[redacted] $f(x) = 2x$ [redacted] $x = 0$ [redacted] $x = 3$[redacted] What is the area under the curve $f(x) = 2x$ between $x = 0$ and $x = 3$?



a. 6

b. 9

c. 12

d. 3

31 [Intermediate] • Target: Average value of a function

[redacted] $y = ax^2 + 1$ [redacted] $x = 0$ [redacted] $x = 1$ [redacted] $\frac{7}{3}$[redacted] a. If the area of the region bounded by the curve $y = ax^2 + 1$, the x-axis, and the lines $x = 0$ and $x = 1$ is equal to $\frac{7}{3}$, find the value of the constant a.

a. 2

b. 3

c. 1

d. 4

32 [Intermediate] • Target: Word problems: integration in context

[redacted] $y = x^1$ [redacted] $x = 0$ [redacted] $x = 4$[redacted] Find the area under the curve $y = x^1$ from $x = 0$ to $x = 4$.

a. 4

b. 6

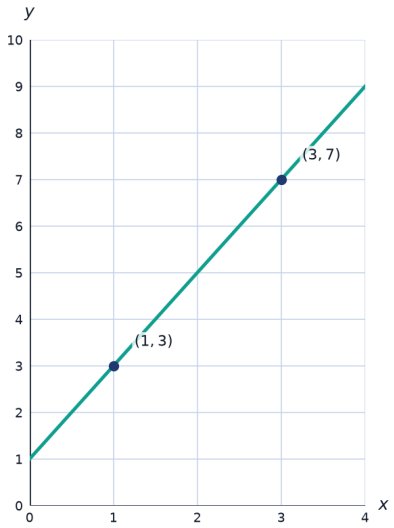
c. 5

d. 8

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33 **[Intermediate]** • **Target: MCQ — Basic antiderivatives (the power rule)**

[redacted] $f(x) = 2x + 1$ [redacted] $x = 1$ [redacted] $x = 3$. Calculate the definite integral of the function $f(x) = 2x + 1$ from $x = 1$ to $x = 3$.



a. 8

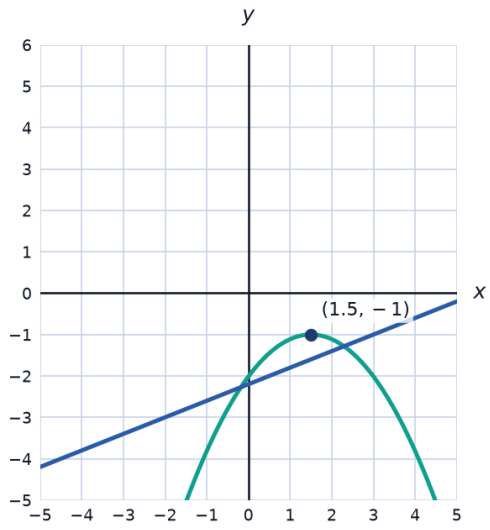
b. 6

c. 10

d. 12

34 **[Intermediate]** • **Target: MCQ — Antiderivatives of polynomials**

[redacted] $f(x) = x^2 - 6x + k$ [redacted] $x = 3$. [redacted] The function $f(x) = x^2 - 6x + k$ represents a parabola that touches the x-axis at $x = 3$. What is the value of k?



a. 9

b. -9

c. 0

d. 3

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35 [Intermediate] • Target: MCQ — Antiderivatives of trigonometric and exponential functions

[redacted] $f(t) = -5t^2 + 20t + 1$ [redacted] ([redacted]) [redacted] t [redacted]. [redacted] A particle is projected vertically upwards. The function $f(t) = -5t^2 + 20t + 1$ represents the height (in meters) of the particle after t seconds. What is the maximum height reached by the particle?

a. 15 [redacted]

b. 21 [redacted]

c. 25 [redacted]

d. 16 [redacted]

36 [Intermediate] • Target: MCQ — Evaluating definite integrals

[redacted] $f(x) = a \cdot e^{kx}$[redacted] $(0, 5)$ [redacted] $(\ln(3), 15)$. [redacted] a [redacted] k. For the exponential function $f(x) = a \cdot e^{kx}$, two points on its graph are $(0, 5)$ and $(\ln(3), 15)$. Find the value of the constant a and the value of the constant k.

a. $a = 5, k = \ln(3)$

b. $a = 5, k = 3$

c. $a = 5, k = \frac{\ln(3)}{1}$

d. $a = 5, k = \ln(15/5)$

37 [Intermediate] • Target: MCQ — Definite integrals with polynomial integrands

[redacted] $f(x) = 3x^2 + 2x - 1$[redacted] $\int_0^2 f(x)\,dx$[redacted] If $f(x) = 3x^2 + 2x - 1$, what is the value of the definite integral $\int_0^2 f(x)\,dx$?

a. 14

b. 10

c. 12

d. 8

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: MCQ — Definite integrals of trigonometric and exponential functions • Cross-Lesson: 8.01 Exponentials & Logarithms

[redacted] $f(x) = ax^2 + bx + c$ [redacted] $(2, -5)$ [redacted] $(4, 7)$. [redacted] a[redacted] A quadratic function $f(x) = ax^2 + bx + c$ has its vertex at $(2, -5)$ and passes through the point $(4, 7)$. What is the value of a?

a. 2

b. 3

c. 4

d. 1

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39 [Challenging] • Target: MCQ — Area under a curve • Cross-Lesson: 9.01 Statistics & Probability

■■ ■■■■ ■■■■■■■■ ■■■■■■■: $\int_{-2}^{2}(x^3+x)\,dx$■ Integral of odd function on symmetric interval?

a. 0

b. 4

c. 8

d. -4

40 [Challenging] • Target: MCQ — Signed area and regions below the axis • Cross-Lesson: 1.01 Algebra & Equations

■■ ■■■■ ■■■■■■■■ ■■■■■■■: $\int_{a}^{a} f(x)\,dx$ ■■■ ■■■■ $f(x)$■ Definite integral with identical upper and lower limits?

a. 0

b. 1

c. $f(a)$

d. $2a$

41 [Challenging] • Target: MCQ — Properties of definite integrals • Cross-Lesson: 2.01 Functions & Transformations

■■ ■■■■ ■■■■■■■■ ■■■■■■■: $\int_{1}^{2}\frac{2x+1}{x^2+x}\,dx$■ Evaluate integral where numerator is derivative of denominator.

a. $\ln 3$

b. $\ln 6$

c. $\ln 2$

d. 1

42 [Challenging] • Target: MCQ — Finding a function from its derivative • Cross-Lesson: 3.01 Sequences & Series

■■ ■■■■ ■■■■■■■■: $\int\frac{2x}{x^2+1}\,dx$■ Evaluate integral where numerator is derivative of denominator.

a. $\ln(x^2+1)+C$

b. $\frac{1}{x^2+1}+C$

c. $\arctan(x)+C$

d. $2\ln(x^2+1)+C$

43 [Challenging] • Target: MCQ — Average value of a function • Cross-Lesson: 4.01 Trigonometry

■■■ ■■■■ $f(x)$ ■■■■ ■■■■■ ■■■■ $\int_{0}^{3} f(x)\,dx = 5$■ ■■■ ■■■■ $\int_{-3}^{3} f(x)\,dx$■ Integral of even function on symmetric interval?

a. 10

b. 0

c. 5

d. -5

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44 [Challenging] • Target: MCQ — Word problems: integration in context • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

■■■ ■■■ $\int_1^4 f(x)\,dx = 8$■ ■■■ ■■■■ $\int_1^4 3f(x)\,dx$■ Constant multiple property of integrals?

a. 24

b. 8

c. 11

d. 12

45 [Challenging] • Target: Basic antiderivatives (the power rule) • Cross-Lesson: 6.01 Limits & Derivatives

■■ ■■■■ ■■■■■■■ ■■■■■■: $\int_0^3 2x\,dx$■ Evaluate definite integral $\int_0^3 2x dx$.

a. 9

b. 6

c. 3

d. 18

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